The Significance of Body Surface Area to Mass Ratio for Thermal Responses to a Standardized Exercise-Heat Stress Test

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ABSTRACT

AKAVIAN, I., Y. EPSTEIN, A. RABOTIN, S. PERETZ, N. CHARKOUDIAN, and I. KETKO. The Significance of Body Surface Area to Mass Ratio for Thermal Responses to a Standardized Exercise-Heat Stress Test. *Med. Sci. Sports Exerc.*, Vol. 57, No. 1, pp. 88–93, 2025. **Purpose:** To evaluate the significance of body surface area-to-mass ratio (BSA/mass) on the heat-tolerance test (HTT) results. We hypothesized that individuals defined as heat tolerant (HT) would have on average higher BSA/mass compared with heat intolerant (HI) individuals. **Methods:** A retrospective reanalysis of the HTT results of 517 soldiers (age, 18–38 yr; M/F, 96%:4%), who were tested by the Israel Defense Forces HTT protocol. The criterion for heat tolerance in the current analysis was a rectal temperature (T_{re}) plateau during the second hour of the test. A logistic regression analysis to evaluate the predictive power of BSA/mass for heat intolerance was performed; the spline model was applied to show the odds for heat intolerance across BSA/mass. **Results:** In men BSA/mass of HI individuals was lower than HT individuals (248 ± 19 vs 262 ± 18 $cm^2 \cdot kg^{-1}$, $P < 0.01$, d = 0.76). In women a similar trend was noted but with no statistical significance between HT and HI groups. The odd ratio for heat intolerance for every unit increase in BSA/mass was 0.97 (95% confidence interval, 0.95–0.99). The spline model plateaued above BSA/mass of 270 $cm^2 \cdot kg^{-1}$. **Conclusions:** The results imply that body-core temperature responses to a standard exercise-heat stress (fixed external work rate and climatic conditions) are influenced by BSA/mass. More specifically, lack of a steady state in T_{re} (indicating heat intolerance) was more likely to occur with every unit decrease in BSA/mass. These findings contribute to a better understanding of the role of body anthropometry in the response to a standard exercise-heat task that might have an implication on clinical decision making about return to duty/play of soldiers, athletes, and others who deemed to be identified as HI. **Key Words:** HEAT TOLERANCE, HEAT STRESS, ANTHROPOMETRY, THERMAL STRAIN

Exertional heat illnesses (EHI) can be viewed as a continuum of syndromes resulting from physical activity, typically, but not necessarily, in the heat. Exertional heat illnesses range from heat exhaustion (least serious) to heat injury and exertional heat stroke (EHS), the most serious (and sometimes fatal) condition (1). In general, EHI is characterized by elevated body-core temperature; in the case of EHS body temperature might be extremely high (>40°C) and is accompanied, at its acute stage, by altered mental status and later by multiorgan failure (2).

During exercise-heat stress, body core temperature will rise, reflecting an imbalance between heat gain and heat dissipation. Although some heat is gained from the environment, depending on the temperature gradient between the environment and the skin, most of the heat accumulated in the body during exercise is the product of the metabolic rate that is generated by active skeletal muscle. On the other hand, heat is dissipated mostly by sweat evaporation in proportion to body surface area (BSA) and is dependent on the vapor pressure gradient between the skin and the environment (3). Consequently, individuals with a larger BSA to mass ratio (BSA/mass), have a higher cooling potential by sweat evaporation than individuals with smaller BSA/mass (4,5).

Under field conditions, there are some caveats to the assumption that high BSA/mass is always beneficial for heat dissipation, including contributions of adiposity, muscle mass and other insulative factors (e.g., clothing). Nonetheless, Taylor et al. recently showed a strong inverse relationship between BSA/mass and risk of exertional heat illness in a group of 4600 active duty servicemembers (744 EHS and 3720 controls), presenting an important practical contribution of this relationship (6).

A heat tolerance test (HTT) is commonly used to obtain quantitative information regarding an individual's ability to tolerate exercise-heat stress (7–9). This test is often used as part of the decision on return to duty or play (RTD/P) for individuals who

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recovered from exertional heat illness (9–13). There are several HTT protocols that have been published (9,12–14). In most of the cases, those tests, including the Israel Defense Forces (IDF) HTT, are aimed to assess the individual's thermoregulatory system functionality under heat stress. In general, these do not serve as a diagnostic test, or a task-oriented test (11). In a review of current HTT concepts, Buttler and colleagues evaluated the IDF HTT as the only test graded as "A," signifying strong evidence for its effectiveness (8).

In a recent case report describing a Special Forces soldier who failed the IDF-HTT twice, it was suggested that the underlying cause was a low BSA/mass (15). Our goal in the present study was to investigate this possibility and evaluate in a large cohort the contribution of BSA/mass to thermal responses during the exercise-heat stress of the IDF-HTT. We hypothesized that individuals classified "heat intolerant" (HI) would, on average, have a lower BSA/mass compared with those classified "heat tolerant" (HT).

METHODS

Setting and Participants. This study focuses on the analysis of post-EHI cases of active duty IDF soldiers who performed the 120 min IDF-HTT between the years 2007 and 2021. We utilized the retrospective HTT registry for our data collection. The IDF Medical Corps Institutional Review Board approved this study and waived the requirement for informed consent based on strict protection of the anonymity of the individuals included (1198-2012-IDF).

The Heat Tolerance Test (HTT). The IDF-HTT is typically conducted within 2 to 6 wk after the recovery from the EHI event. Preceding the HTT, subjects undergo a comprehensive medical evaluation, and their weight, height, and fat percentage are measured. Body fat percentage is assessed utilizing the Durnin and Womersley four-site skinfold equation (15). Body surface area is computed according to Dubois and Dubois formula (16).

The participant, dressed in shorts, walks on a treadmill for 2 h at 5 $km \cdot h^{-1}$ (1.39 $m \cdot s^{-1}$) with a 2% incline in an environmental chamber set at 40°C and 40% relative humidity (RH), with no solar radiation (14). For practical reasons the test does not consider time of the year (acclimatization) nor physical fitness as covariant. Core body temperature (rectal temperature [T_{re}]) and heart rate (HR) are continuously monitored throughout the test. Sweat rate is calculated from post to pre weight difference (±10 g) corrected for fluid intake and urine output. Several criteria assist in defining a negative/positive test (=heat tolerance/intolerance) (14,17,18). It is noteworthy that, when an individual fails the test, a second test is scheduled 3 months later to assess potential changes in heat tolerance. Only if the second (and occasionally the third) test is also positive the individual is defined as HI and his or her medical profile is amended accordingly.

HTT Analysis and Statistical Methods. Subjects who were defined as HI were compared with those who were HT. Maintaining a plateau in T_{re} (≤38.5°C) during the second hour of the test was the criterion used in the present study to define heat tolerance; a plateau was considered when the delta Tre between 60 and 120 min of the test was lower than 0.45°C (18,19). If an examinee took more than one test, then the last test was taken for data analysis. Categorical variables are presented as frequencies (percentage), and continuous variables as mean (± standard deviation [SD]). For continuous variables —age, BSA, BSA/mass, BMI and fat percentage, we applied *t* test to compare between the HI and HT groups. The effect of anthropometric variables on heat tolerance/intolerance was tested separately for men and women. Due to the small sample size in the women's subgroup, we confirmed normal distribution by employing the Shapiro-Wilk normality test ($P > 0.05$). The Cohen's effect size (d) was used to measure the strength of the relationship between the different parameters and tolerance to heat, whenever it was found to be statistically significant.

To analyze the relationship between BSA/mass ratio and heat intolerance, logistic regression analysis unadjusted and adjusted for age as a potential confounder, was applied. This analysis modeled the probability of a binary outcome—heat intolerance/heat tolerance, based on the predictor variable —BSA/mass ratio. An odds ratio being greater than 1 indicates that higher BSA/mass values are associated with increased odds of heat intolerance, whereas an odds ratio less than 1 suggests decreased odds.

Based on the logistic regression analysis, a spline model was used to explore the relationship between BSA/mass ratio and heat intolerance. The spline model allows the examination of potential nonlinear relationships by utilizing piecewise polynomial functions connected at specific points, or knots. By incorporating 4 degrees of freedom and 95% confidence intervals, the spline model captured the curvature in the relationship between BSA/mass and the odds of heat intolerance, providing a flexible and robust approach to visualizing this association.

All statistical analyses were performed using R software, version 2023.06.2 (R Core Team, Vienna, Austria), with a significance level set at $\alpha = 0.05$.

RESULTS

Demographic Characteristics. A total of 517 soldiers were included in the analysis (M/F, 96%:4%). The mean age of the participants was 19.8 yr (SD = 2.1 yr), ranging from 17 to 38 yr. Figure 1 shows the study population selection process and the heat intolerance distribution across subsequent HTTs. Overall, in the first HTT, 16% of the examinees were diagnosed as HI, and after repeated tests (up to three in total) 6.7% of the population were deemed HI. A total of 13.6% (3 / 22) of the women were diagnosed as HI, compared to 6.4% (32 / 492) of the men. Table 1a and 1b shows the subgroup analysis for men and women separately.

Anthropometric Characteristics. Heat intolerant male individuals had higher body weight (79 ± 13 vs 73 ± 11 kg, $P < 0.02$, d = 0.50), higher fat percentage (20.1 ± 6.1% vs 16.5 ± 4.4%, $P < 0.01$, d = 0.69), and lower BSA/mass

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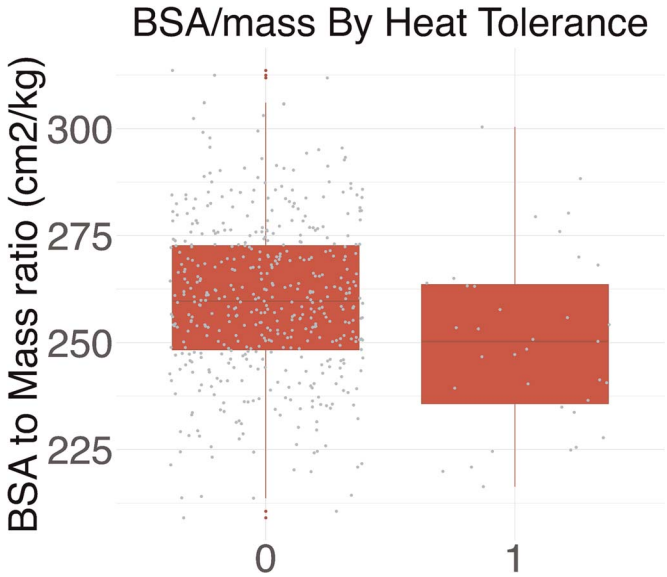

FIGURE 2—BSA to mass ratio ($cm^2 \cdot kg^{-1}$) of heat intolerant vs heat tolerant subjects. This figure illustrates the distribution of BSA to mass ratio among heat-tolerant (0) and heat-intolerant (1) individuals using a boxplot graph with superimposed individual data points.

exercise in the heat. In the present analysis, the logistic regression model showed that the odds for heat intolerance decreased with every unit increase in BSA/mass. In the IDF, heat intolerance results in a delay in RTD and when the medical profile is amended it prohibits the individual from returning to some combat positions. Similar tests and approaches are used in other militaries, athletic organizations, and industrial settings (9–13,20,21).

For men weight, BMI, and percent body fat were found to be significantly different between the HT and HI groups (Table 1a). Nevertheless, the effect size of BSA/mass was much higher than for all other parameters. This suggests that, in terms of anthropometric variables, BSA/mass was the strongest contributor to these thermal responses. For women, no significant differences in any anthropometric variables were found between the two groups (for further discussion on sex differences see below).

Considerations for Return to Duty/Play. The common procedure in the IDF is to schedule an HTT 2 to 6 wk after recovery from EHI. In the case of a positive test result (HI), a second test is scheduled 3 months later. In the present cohort, the prevalence of HI was 16% during the first test, which decreased to 6.7% in a subsequent test. This is in accord with a previous observation, where HI prevalence dropped from 10% to 2%) (22) with repeated tests. This observation also supports an existing hypothesis that after EHI, and especially EHS, there might be a period of transient heat intolerance, which improves over time (23–25). This observation has implications for the formulation of RTD/P guidelines after recovery from EHI, emphasizing the importance of accounting for the time needed for regaining physiological adaptations after an EHI episode.

In many cases, the underlying cause for failing the HTT and defining an individual as HI is clear. High percentage of body fat

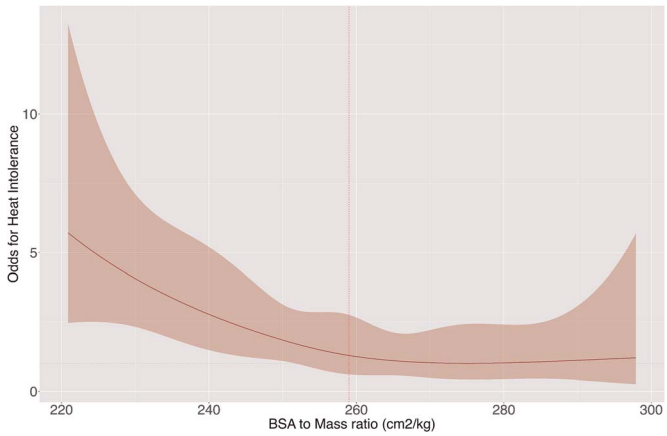

FIGURE 3—Spline curve depicting the odds for heat intolerance according to BSA/M ($cm^2 \cdot kg^{-1}$), using logistic regression analysis (adjusted for age). Spline model is presented for the odds ratios (ORs) and 95% CI for HI by BSA/M ($cm^2 \cdot kg^{-1}$). The model was adjusted for age. The line represents the OR and the shaded areas indicate the 95% CI. Dashed gray line set at OR = 1, dashed red line set at average BSA/mass = 259 $cm^2 \cdot kg^{-1}$.

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and very low physical fitness are among the most possible causes, beyond congenital background (e.g., idiopathic anhidrosis) (11,25). However, in some cases, there is no straightforward clinical or physiological explanation for a positive HTT, especially among young and healthy soldiers who were found to be suitable for combat roles. Such is the case, for example, with a Special Forces soldier who failed the test twice, which ultimately prevented him to return to active duty as a warfighter. After eliminating all other possible causes, this soldier's low BSA/mass was proposed as the potential cause for failing the test (26). This explanation can be substantiated by several studies that have mentioned BSA/mass regarding the HTT (11,27–30).

Physiology and Biophysics of Heat Dissipation during the HTT. The ability of the body to attain thermal equilibrium (i.e., a steady-state in body-core temperature) depends on the ability to match heat dissipation to heat gain. Under the conditions of the HTT heat gain from the surrounding is low (calculated, <70 W) and metabolic heat production is the major source of heat accumulated in the body. However, given the low metabolic rate (calculated, 320–350 W, depending on body weight), metabolic heat production alone is unlikely to be the major contributor to results of the test. In general, in the young active population, in contrast to the elderly, peripheral vasodilation and sweat secretion are not compromised and are not considered a limiting factor for heat dissipation (31). We do not have data on peripheral vasodilation, but whole body sweat rate was similar in both groups, suggesting that hypohidrosis is not a primary cause of HI in young healthy people. This is in accord with a previous smaller study (comprising 9 heat-intolerant subjects and 9 controls), which demonstrated that HI individuals exhibited sweat rates and heat conductance similar to that of HT subjects, but those in the HI group displayed a lower BSA/mass ratio (28). For these reasons, we hypothesized that in the context of the IDF-HTT, individuals with a low BSA/mass would be at a relative disadvantage regarding the defined criteria for heat tolerance. Similarly, Taylor et al. showed in an epidemiological case-controlled study that lower BSA/mass was associated with increased risk of EHS in a healthy group of men and women serving as U.S soldiers (6). Our present data suggest that low BSA/mass is a significant contributor to the results of the IDF-HTT because, although overall sweating rate was similar, the relative area for evaporation was less in those individuals.

Whether other underlying conditions can explain the differences between HT and HI individuals, in this large cohort, cannot be ruled out and deserve further studies. For example, in an earlier study, we have shown that the cardiovascular reserve of HI individuals is lower than that of HT subjects (32). This might imply that impaired CV reserve might underlie heat intolerance.

Consideration of Sex as a Biological Variable. Although women represent increasing percentages of military populations around the world, these groups are still mostly made up of men. In the present data set, men constituted 96% of the cohort, with only 22 women soldiers included. In this small group, the average BSA/mass of the 3 female soldiers who failed the test (13.6%), was low, with no statistical significance between the HT and HI women. Even though the sample size of women was low, the results may be relevant in the context of heat tolerance testing. Furthermore, these results are relevant in terms of the importance of inclusion of sex as a biological variable in human subjects' research (33,34). With the gradual process of integrating women into combat roles, further research is warranted to better address anthropometric differences leading to thermoregulatory variations between men and women.

Limitations. Given that the present cohort primarily consists of young soldiers (mean age, 19.8 yr), there might be a selection bias that restricts the generalizability of our findings to other demographic groups. Furthermore, the relationships shown in the present study might be different for other scenarios – for example, in athletic situations or tests where uniforms and/or environmental conditions result in different thermal and vapor pressure gradient over most of the skin surface (e.g., American football) (35). It should also be noted that for practical reasons the test does not control for physical fitness. From earlier experience, the work intensity is 30% to 40% $\dot{V}O_{2max}$ (36), which for an average fit individual should not be a limiting factor for exhaustion nor explain differences in metabolic heat production between the HT and HI groups. Still, it is important to highlight the unique aspect of our study, which lies in a large cohort and its comprehensive assessment spanning more than a decade, covering each exertional heat injury and HTT. This extensive data set serves as a solid foundation for conducting a large-scale exploration into potential anthropometric contributors to heat intolerance in human subjects.

CONCLUSIONS

In this study, we showed a significant relationship between heat intolerance as defined by the IDF-HTT and BSA/mass. Specifically, the odds for HI decreased with every unit increase in BSA/mass (odds ratio, 0.97; 95% CI, 0.95–0.99) and plateaued above BSA/mass ratio of 270 cm/kg^2. This association may help identify individuals at increased risk for EHS and have implications for the RTD/RTP process for soldiers, athletes or others post EHS. In addition, these findings contribute to a better understanding of the role of body anthropometry as a contributor to physiological responses to exercise in the heat. Finally, the data highlight the complexity of the decision on RTD after recovery from EHI. Any decision on RTD should consider not only the HTT results but also the individual's history and planned activities, whether they be military, athletic, or other endeavors.

This study has no funding sources. No conflicts of interest, financial or otherwise, are declared by the authors. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The outcomes of the present study should be understood as independent research findings and should not be interpreted as an endorsement or approval by the American College of Sports Medicine. N.C. is an employee of the Department of the Army of the United States. The opinions or assertions contained herein are the private views of the author(s) and are not to be construed as official or as reflecting the views of the U.S. Army or the Department of Defense. Citations of commercial organizations and trade names in this report do not constitute an official Department of the Army endorsement or approval of the products or services of these organizations.

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